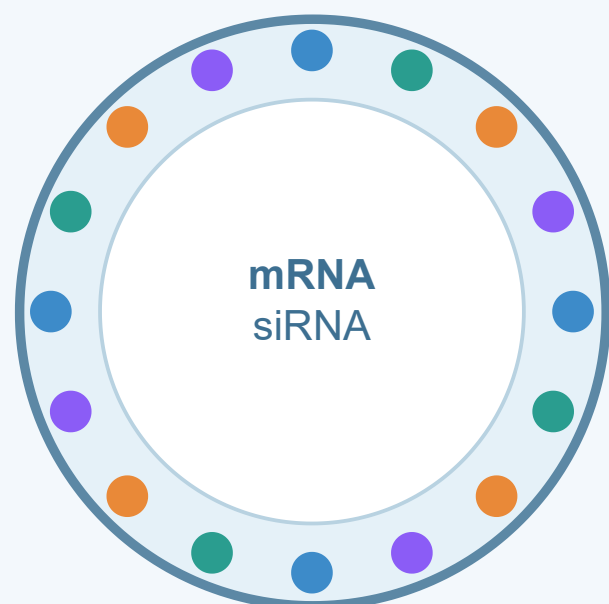


a Formulation and context



Four lipid roles + nucleic-acid cargo



Ionizable · Helper · Sterol · PEG-lipid
molar fractions · role masks · structural support



Target /
tissue



Cargo / route



Mixing /
dose

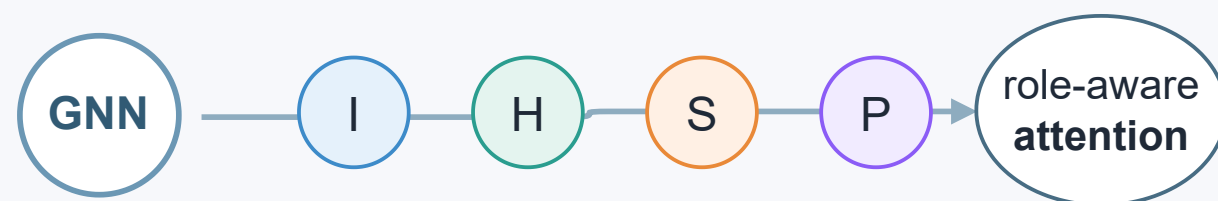
Observed support is explicit

identity · graph · activity are tracked separately

b Dual-path representation

Role-aware molecular graph path

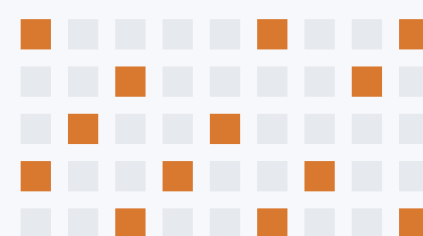
shared GNN · role/ratio tokens · masked attention



Graph features and process context form the point-estimation state.

Evidence-aligned ranking path

ionizable count-Morgan + descriptors + formulation context



MLP
direct rank
evidence

+

A gated residual augments the graph ranking score without changing the point head.

c Prediction heads

Point estimate

heteroscedastic delivery
mean

Screen-local rank

candidate ordering and
top-decile recovery

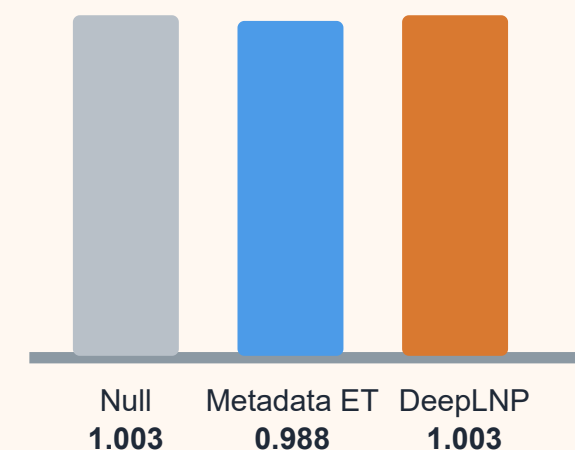


Masked auxiliary heads
size · PDI · zeta
encapsulation · censored
toxicity

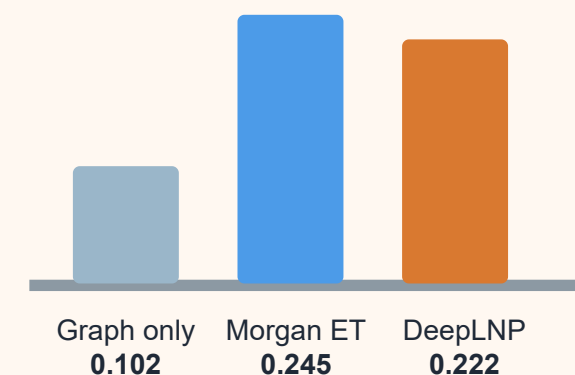
d Held-screen evidence

Five endpoint-balanced folds
38 held-out experimental screens

Point error (RMSE ↓)



Within-screen Spearman (↑)



Separate decisions, separate outputs
absolute error and candidate ranking are
evaluated independently